

TEZPUR UNIVERSITY
Spring Semester End Examination, 2022
CO218: DATA COMMUNICATION

TU/CSE

Full Marks: 60

Time: 3 hrs

All questions are compulsory. Numbers in right hand side indicate marks. Write the answers precisely.

1. Answer the following to the point? (Answer any **five**) [2x5=10]
 - a. Write the difference between transparent and source routing bridge
 - b. Write briefly about 802.1Q and 802.2
 - c. Why is CSMA/CD not applicable in wireless medium?
 - d. Name two encapsulated forms of PPP for DLC
 - e. Write the difference between IEEE 802.4 and IEEE 802.5
 - f. Write briefly about hidden terminal and exposed terminal problems
2. In what situation the collision is still possible in CSMA/CD and how to overcome from such situation? [3]
3. What is a contention system? How does slotting improve performance of ALOHA? [1+2=3]
4. Why OFDM is called multi-carrier modulation? Write two advantages of OFDM technique. [2+1=3]
5. Calculate the polynomial code checksummed frame for a frame 1101011011 using the generator polynomial $x^4 + x^2 + 1$. [4]
6. What is the working principle of *truncated binary exponential backoff (BEF)* algorithm used in CSMA? [4]
7. How does HDLC handle the situations of i) busy condition and ii) time-out recovery? [3+3=6]
8. Consider that carrier frequency of a signal is 250 KHz. If the MFSK is used to encode the data signal with frequency difference by 25 KHz each with 3 bits per signal element, calculate the possible frequency assignments for each of the eight possible 3-bit data combinations. [4]
9. How are flow and error control done in TDM link? Write briefly about why the statistical TDM is important? [4]
10. A time division multiplexer combines four 100-kbps channels using a time slot of 2 bits. Compute the frame rate, frame duration and the bit rate, bit duration? [4]
11. (a) Consider a stream of binary 1s is required to be transmitted with a bit duration of 1 μ s each using NRZ-1 and Manchester encoding. If the 2 number of bits per signal element needed to be encoded, calculate the modulation rates for both the schemes. [5]

Or

(b) Discuss about the working principle of distance vector routing and link state routing algorithms. [5]
12. Consider four digital data channels each transmitting at 1 Mbps speed. Design an appropriate configuration for FDM, using a satellite channel of bandwidth 1 MHz. [4]
13. Draw the waveform to encode the data string "010001010110000000011" for the following encoding schemes (Answer any **three**) [2x3=6]
 - a. Differential Manchester
 - b. B8ZS
 - c. ASK
 - d. DBPSK (take any phase with your convenience)
 - e. Bipolar-AMI